Administración actuarial de riesgos financieros

Tarea larga 1

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Tarea larga 1

Esta tarea se entrega en equipos conformados por a lo más cinco personas; y se entrega a más tardar las **23:59 horas del primero de marzo de 2026 DC**, a la dirección francisco.garcia@udlap.mx

El día siguiente será aplicado el **primer examen parcial**, a la hora de la clase.

El lunes siguiente, **9 de marzo**, a partir de las 10 am y hasta las 14:00 horas, deberán presentarse con su equipo y el profesor le hará una pregunta a cada uno. Solo ese día. Por cada pregunta mal contestada, la evaluación de esa persona será al 80%.

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El documento principal –Word, Power Point o PDF– deben contener Carátula, Cuerpo, Conclusiones y Bibliografía; y deben ir adjuntos los documentos de trabajo –Excel– con fórmulas.

También es objeto de **calificación la ortografía y la redacción** de su entregable (hasta un -25%).

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Los temas a evaluar son:

- Introducción a la administración actuarial de riesgos
- Administración del riesgo de mercado, mediante:
  - Simulación histórica
  - Simulación de Monte Carlo
  - Método paramétrico
  - VaR estresado, y
  - Medidas de riesgo

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Así entonces, la ponderación de los ejercicios es:

Problema	Puntos	Acumulado
I	5	5
II y III	2 puntos c/u	9
IV	10	19
V	20	39
VI	3	42
VII	12 (3 puntos cada problema)	54
VIII y IX	3 puntos c/u	60
X	9 (3 puntos cada problema)	69
XI	10	79
XII	21	100

Tarea larga 1

I. Explique ampliamente:

a) Qué son los riesgos financieros, así como el proceso de la administración de riesgos financieros.

b) Descargue los estados financieros de una compañía de seguros, de un banco y de una casa de bolsa (**de los tres**). Identifique y explique el riesgo de mercado, el riesgo de crédito y el riesgo de liquidez del balance y del estado de resultados de dichas empresas.

c) De la Circular Única de Bancos, la Circular Única de Seguros y Fianzas o de la Circular Única de Casas de Bolsa (la que usted elija), identifique todos los artículos que señalan la administración del riesgo de mercado, y resáltelos como documento adjunto al documento a entregar.

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II. Realice una línea de tiempo que ilustre la implementación de los requerimientos de capital con base en los diferentes riesgos para los bancos con actividad internacional emitidos por el Comité de Supervisión Bancaria de Basilea.

III. Realice un ensayo sobre el Capítulo 1 del libro "*Essentials of risk management*"; y de las siguientes películas: *Margin Call*; *The Corporation* e *Inside Job* (de las tres).

Ojo: El ensayo es un tipo de texto en prosa que explora, analiza, interpreta o evalúa un tema. Se considera un género literario comprendido dentro del género didáctico. Por lo anterior, no quiere un resumen, sino una opinión o un análisis del capítulo.

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IV. Usted es un trader de una tesorería en ***México***. Construya un portafolio de cuatro empresas mexicanas, donde dos coticen en México y las otras dos fuera de México –por ejemplo, en la NYSE o en la LSE–.

k	Valor del Portafolio (Miles de MXN)
1	3,500
2	9,500
3	10,000
4	5,500
	28,500

a) Construya una base de datos limpia de más de 500 observaciones con los precios diarios de los activos de su portafolio en la moneda reportada.

b) Grafique el precio de los activos en el tiempo, así como el de los rendimientos. Realizar una tabla con las estadísticas descriptivas de los rendimientos diarios. Grafique. Interprete.

c) Utilice todos los modelos para determinar el VaR a un día y el C-VaR a 1 día al 97.5% de confianza para el portafolio. Interprete.

d) Utilice todos los modelos para determine el VaR a 10 días al 97.5% de confianza para el portafolio. Interprete.

e) Utilice todos los modelos para obtener el VaR estresado. Interprete.

f) Obtenga el nivel de capital regulatorio requerido para este portafolio, previo a la ponderación de riesgos. Interprete.

g) Haga una presentación que ilustre su portafolio [¿por qué decidieron estos activos?], precios de activos, histogramas y presente sus resultados en una tabla resumen. Explique.

Adjuntar como objeto su archivo de trabajo de Excel, muestre todos sus cálculos para llegar al resultado y resáltelo en el Excel —no quiero cifras en valores, muestre sus fórmulas—.

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V. Usted es el *Chief Risk Officer* (CRO) de una organización suiza, país donde labora. El área de trading le informa que se tienen en prospectación los siguientes dos portafolios en instrumentos de capital en los que se desea invertir, conformados por los siguientes activos:

K	Activo	Valor del Portafolio Millones
1	Niikei 225 (Bolsa de Tokio)	15
2	Hang Seng Index (Bolsa de Hong Kong)	20
3	Shenzhen Index (Bolsa de China)	25
4	S&P/ASX200 (Bolsa de Australia)	40
Total		100

K	Activo	Valor del Portafolio Millones
1	S&P500	15
2	DAX	20
3	FTSE 100	25
4	NIKKEI 225	40
Total		100

Cifras en **francos suizos** al XXX de 2026.

Obtenga la información sobre los precios diarios de estos activos y responda las siguientes preguntas.

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a) Construya una base de datos con los precios en la moneda reportada.

b) Grafique el precio de los activos en el tiempo, así como el de los rendimientos. Realizar una tabla con las estadísticas descriptivas de los rendimientos diarios. Grafique. Interprete.

c) Obtenga el VaR (a 1 día y a 10 días), el CVaR (expected shortfall) y el capital por riesgo de mercado (al 97.5% de confianza) de los dos portafolios mediante los siguientes métodos (interprete):

   - Simulación histórica
   - Simulación de Monte Carlo
   - Método paramétrico

Adjuntar como objeto su archivo de trabajo de Excel, muestre todos sus cálculos para llegar al resultado y resáltelo en el Excel —no quiero cifras en valores, muestre sus fórmulas—.

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d) Obtenga el VaR estresado (a 1 día y a 10 días), el CVaR estresado (expected shortfall) y el capital por riesgo de mercado con base en el VaR estresado (al 97.5% de confianza) de los dos portafolios anteriores con los siguientes métodos (interprete):

- Simulación histórica
- Simulación de Monte Carlo
- Método paramétrico

Adjuntar como objeto su archivo de trabajo de Excel, muestre todos sus cálculos para llegar al resultado y resáltelo en el Excel —no quiero cifras en valores, muestre sus fórmulas—.

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d) En calidad de CRO, debe presentar todos los resultados que obtuvo ante el **Comité de Riesgos y ante el Comité de Inversiones**, así como al CEO y al CFO el día siguiente por la mañana.

Por lo que, con base en sus resultados obtenidos, de las volatilidades encontradas en sus activos, del **entorno económico y político** de los países donde operan los activos, interprete sus resultados de forma muy ejecutiva y preséntelos en un archivo PPT en forma de bullets (de requerirlo, presente gráficas), y recomiende **cuál de los dos portafolios** es el que la organización debe mantener.

Explique ampliamente su decisión.

Adjunte su archivo PPT como objeto.

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VI. En clase se describió el Modelo Geométrico Browniano para modelar el precio de los activos:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Explique con palabras ***no técnicas*** los siguientes componentes de dicha ecuación:

- dS_t
- $\mu S_t dt$
- $\sigma S_t dW_t$
- $\mu S_t dt + \sigma S_t dW_t$

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VII. En clase se describió el Lema de Ito para determinar el precio futuro de los activos con riesgo. El modelo resultante fue el siguiente:

$$S_T = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}\varepsilon_t}$$

a. Explique con palabras ***no técnicas*** cada uno de los componentes de esta ecuación.

- S_T
- $\left(\mu - \frac{\sigma^2}{2}\right)(T-t)$
- $\sigma\sqrt{T-t}\varepsilon_t$
- $e^{\left(\mu - \frac{\sigma^2}{2}\right)(T-t) + \sigma\sqrt{T-t}\varepsilon_t}$

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b. Estando en febrero de 2026, realice un **diagrama de flujo** que describa el proceso para estimar el precio de un activo al cierre del año 2026.

c. Estando en febrero de 2026, estime para el cierre del año 2026 el **valor del portafolio** elegido en la pregunta IV.

d. Estando en febrero de 2026, obtenga el VaR y el CVaR al 97.5% de confianza de este portafolio al cierre de diciembre de 2026. Explique ampliamente.

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VIII. ¿Qué significa la métrica del CVaR, y porqué la regulación prefiere esta métrica a la del VaR? Amplíe su respuesta.

IX. Ofrezca un ejemplo de **grandes pérdidas financieras** observadas en la evidencia empírica por **riesgo de mercado**, resaltando la falla en sus controles en la administración de dicho riesgo, los protagonistas y una propuesta de solución. De ser usted el **CRO**, ¿qué habría hecho diferente?

X. Sin saltarse pasos, y quitando los ejercicios de tarea corta y los de ejemplos, realice **tres ejercicios** de los que están al final del **Capítulo IV.** Medidas de riesgo del libro *Nonlife actuarial models. Theory, methods and evaluation.*

XI. Explique e ilustre con un ejemplo que el VaR no es una medida coherente de riesgo; pero el CVaR sí.

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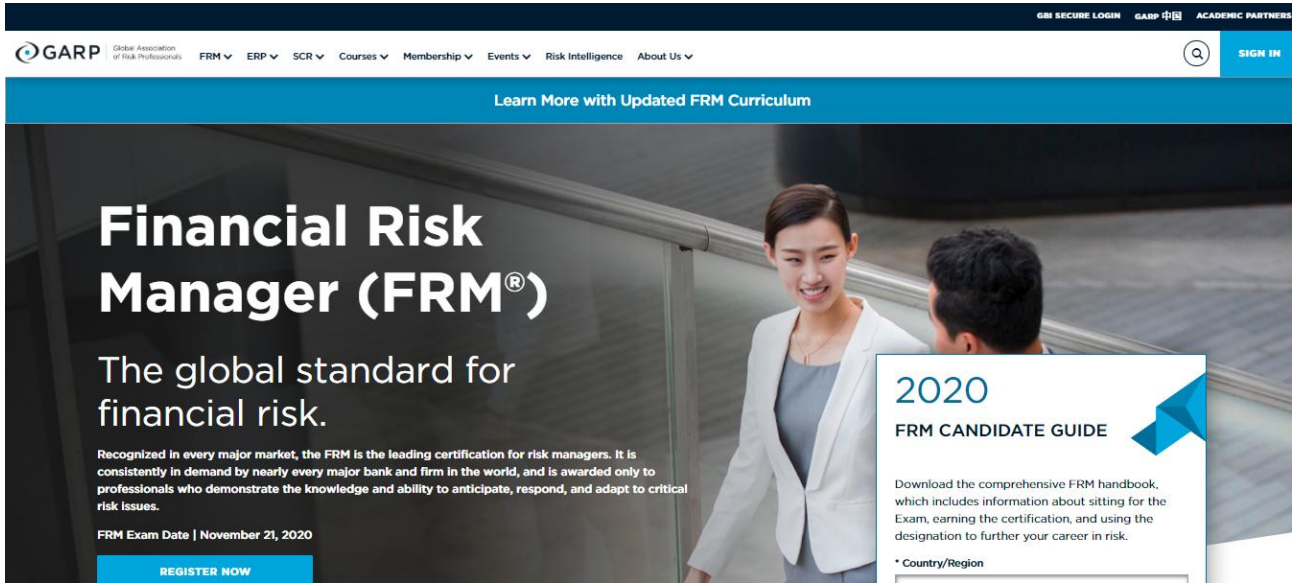
XII. La organización suiza para la que trabaja le está pidiendo que se certifique con el *Financial Risk Management* de la *Global Association of Risk Professionals*, *GARP*. Por lo que se pone a estudiar arduamente en prepararse para ello. Así, resuelva los siguientes ejercicios que son tomados de exámenes anteriores, y ***explique su respuesta ampliamente***.



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EXAMPLE 5.2: INDEPENDENCE

A fundamental assumption of the random walk hypothesis of market returns is that returns from one time period to the next are statistically independent. This assumption implies

a. Returns from one time period to the next can never be equal.

b. Returns from one time period to the next are uncorrelated.

c. Knowledge of the returns from one time period does not help in predicting returns from the next time period.

d. Both b. and c. are true.

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Example 16-3a: FRM Exam 1999----Question 83

In the presence of fat tails in the distribution of returns, VAR based on the delta-normal method would (for a linear portfolio)

a. Underestimate the true VAR

b. Be the same as the true VAR

c. Overestimate the true VAR

d. Cannot be determined from the information provided

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FRM Exam 1999 Question 89/Market Risk. What is the correct interpretation of a $3 million overnight VAR figure with 99% confidence level? The institution

a) Can be expected to lose at most $3 million in 1 out of next 100 days

b) Can be expected to lose at least $3 million in 95 out of next 100 days

c) Can be expected to lose at least $3 million in 1 out of next 100 days

d) Can be expected to lose at most $6 million in 2 out of next 100 days

Tarea larga 1

FRM Exam 1998 Question 22/Capital Markets. Considering arbitrary portfolios A and B, and their combined portfolio C, which of the following relationships always holds for VARs of A, B, and C?

a) $VaR_A + VaR_B = VaR_C$

b) $VaR_A + VaR_B \geq VaR_C$

c) $VaR_A + VaR_B \leq VaR_C$

d) None of the above

Tarea larga 1

FRM EXAM 2007—QUESTION 50. A portfolio consists of two zero-coupon bonds, each with a current value of $10. The first bond has a modified duration of one year and the second has a modified duration of nine years. The yield curve is flat, and all yields are 5%. Assume all moves of the yield curve are parallel shifts. Given that the daily volatility of the yield is 1%, which of the following is the best estimate of the portfolio's daily value at risk (VAR) at the 95% confidence level?

a) USD 1.65

b) USD 2.33

c) USD 1.16

d) USD 0.82

Tarea larga 1

FRM Exam 1999. What is the correct interpretation of a $3 million overnight VAR figure with 99% confidence level? The institution

a) Can be expected to lose at most $3 million in 1 out of next 100 days.

b) Can be expected to lose at least $3 million in 95 out of next 100 days.

c) Can be expected to lose at least $3 million in 1 out of next 100 days.

d) Can be expected to lose at most $6 million in 2 out of next 100 days.

Tarea larga 1

FRM Exam 2005. The 10-Q report of ABC Bank states that the monthly VaR of ABC Bank is USD 10 million at the 95% confidence level. What is the proper interpretation of this statement?

a) If we collect 100 monthly gain/loss data of ABC Bank, we will always see five months with losses larger than $10 million..

b) There is a 95% probability that the bank will lose less than $10 million over a month.

c) There is a 5% probability that the bank will gain less than $10 million each month.

d) There is a 5% probability that the bank will lose less than $10 million over a month.

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FRM EXAM 2003—QUESTION 11. Based on a 90% confidence level, how many exceptions in back testing a VAR would be expected over a 250-day trading year?

a) 10

b) 15

c) 25

d) 50

Tarea larga 1

FRM Exam 1998. Considering arbitrary portfolios A y B, and their combination C, which of the following relationship always holds for VaRs of A, B, and C?

a) $VaR_A + VaR_B = VaR_C$

b) $VaR_A + VaR_B \geq VaR_C$

c) $VaR_A + VaR_B \leq VaR_C$

d) None

Tarea larga 1

FRM Exam 1997. To convert VaR from a one day holding period to a ten day holding period the VaR number is generally multiplied by

a) 2.33

b) 3.16

c) 7.25

d) 10

Tarea larga 1

FRM Exam 2001. Rank the following portfolios from least risky to most risky. Assume 252 trading days a year and there are 5 trading days per week

Portfolio	VaR	Holding periodo in days	Confidence Interval
1	10	5	99
2	10	5	95
3	10	10	99
4	10	10	95
5	10	15	99
6	10	15	95

a) 5, 3, 6, 1, 4, 2

b) 3, 4, 1, 2, 5, 6

c) 5, 6, 1, 2, 3, 4

d) 2, 1, 5, 6, 4, 3

Tarea larga 1

FRM Exam 1997. Which of the following quantitative standards is not required by the Amendment to the Capital Accord to incorporate market risk?

a) Minimum holding period of 10 days.

b) 99th percentile, one tailed confidence interval

c) Minimal historical observation period of two years

d) Update of data sets at least quarterly

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FRM Exam 2003. Given the following 30 ordered percentage returns of an asset, calculate the VAR and expected shortfall at a 90% confidence level: −16, −14, −10, −7, −7, −5, −4, −4, −4, −3, −1, −1, 0, 0, 0, 1, 2, 2, 4, 6, 7, 8, 9, 11, 12, 12, 14, 18, 21, 23.

a) VaR (90%) = 10, expected shortfall = 14

b) VaR (90%) = 10, expected shortfall = 15

c) VaR (90%) = 14, expected shortfall = 15

d) VaR (90%) = 18, expected shortfall = 22

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FRM Exam 2008. Assume that the P&L distribution of a liquid asset is i.i.d. normally distributed. The position has a one-day VAR at the 95% confidence level of $100,000. Estimate the 10-day VAR of the same position at the 99% confidence level.

a) $1,000,000

b) $450,000

c) $320,000

d) $220,000

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FRM Exam 2009. Assume that portfolio daily returns are independent and identically normally distributed. Sam Neil, a new quantitative analyst, has been asked by the portfolio manager to calculate portfolio VARs over 10, 15, 20, and 25 days. The portfolio manager notices something amiss with Sam's calculations, displayed here. Which one of the following VARs on this portfolio is inconsistent with the others?

a) VAR(10-day) = USD 316M

b) VAR(15-day) = USD 465M

c) VAR(20-day) = USD 537M

d) VAR(25-day) = USD 600M

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FRM Exam 2008. Which of the following statements about stress testing are true?

I. Stress testing can complement VAR estimation in helping risk managers identify crucial vulnerabilities in a portfolio.

II. Stress testing allows users to include scenarios that did not occur in the lookback horizon of the VAR data but are nonetheless possible.

III. A drawback of stress testing is that it is highly subjective.

IV. The inclusion of a large number of scenarios helps management better understand the risk exposure of a portfolio.

a) I and II only.

b) III and IV only.

c) I, II, and III only.

d) I, II, III, and IV.

Tarea larga 1

FRM Exam 2006. Which of the following is true about stress testing?

a) It is used to evaluate the potential impact on portfolio values of unlikely, although plausible, events or movements in a set of financial variables.

b) It is a risk management tool that directly compares predicted results to observed actual results. Predicted values are also compared with historical data.

c) Both a. and b. are true.

d) None of the above are true.

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FRM Exam 2008. John Flag, the manager of a $150 million distressed bond portfolio, conducts stress tests on the portfolio. The portfolio's annualized return is 12%, with an annualized return volatility of 25%. In the past two years, the portfolio encountered several days when the daily value change of the portfolio was more than 3 standard deviations. If the portfolio would suffer a 4-sigma daily event, estimate the change in the value of this portfolio.

a) $9.48 million

b) $23.70 million

c) $37.50 million

d) $150 million

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FRM Exam 2009. An analyst gathered the following information about the return distributions for two portfolios during the same time period:

Portfolio	Skewness	Kurtosis
A	−1.6	1.9
B	0.8	3.2

The analyst states that the distribution for Portfolio A is more peaked than a normal distribution and that the distribution for Portfolio B has a long tail on the left side of the distribution. Which of the following is correct?

a) The analyst's assessment is correct.

b) The analyst's assessment is correct for Portfolio A and incorrect for Portfolio B.

c) The analyst's assessment is not correct for Portfolio A but is correct for Portfolio B.

d) The analyst's assessment is incorrect for both portfolios.

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FRM Exam 2007. Suppose that A and B are random variables, each follows a standard normal distribution, and the covariance between A and B is 0.35. What is the variance of $(3A + 2B)$?

a) 14.47

b) 17.20

c) 9.20

d) 15.10

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FRM Exam 2009. Assume that a random variable follows a normal distribution with a mean of 80 and a standard deviation of 24. What percentage of this distribution is not between 32 and 116?

a) 4.56%

b) 8.96%

c) 13.36%

d) 18.15%

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FRM Exam 1999. Which of the following statements best characterizes the relationship between the normal and lognormal distributions?

a) The lognormal distribution is the logarithm of the normal distribution.

b) If the natural log of the random variable X is lognormally distributed, then X is normally distributed.

c) If X is lognormally distributed, then the natural log of X is normally distributed.

d) The two distributions have nothing to do with one another.

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FRM Exam 1997 Question 7/Risk Measurement. To convert VAR from a one-day holding period to a ten-day holding period the VAR number is generally multiplied by

a) 2.33

b) 3.16

c) 7.25

d) 10.00

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FRM Exam 2007. The skew of a lognormal distribution is always

a) Positive

b) Negative

c) 0

d) 3

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FRM EXAM 2009—QUESTION 1-11. Based on the risk assessment of the CRO, Bank United's CEO decided to make a large investment in a levered portfolio of CDOs. The CRO had estimated that the portfolio had a 1% chance of losing $1 billion or more over one year, a loss that would make the bank insolvent. At the end of the first year the portfolio has lost $2 billion and the bank was closed by regulators. Which of the following statements is correct?

a. The outcome demonstrates a risk management failure because the bank did not eliminate the possibility of financial distress.

b. The outcome demonstrates a risk management failure because the fact that an extremely unlikely outcome occurred means that the probability of the outcome was poorly estimated.

c. The outcome demonstrates a risk management failure because the CRO failed to go to regulators to stop the shutdown.

d. Based on the information provided, one cannot determine whether it was a risk management failure.

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FRM EXAM 2002—QUESTION 126. Consider that a stock price S that follows a geometric Brownian motion $dS = aSdt + bSdz$, with b strictly positive. Which of the following statements is false?

a) If the drift a is positive, the price one year from now will be above today's price.

b) The instantaneous rate of return on the stock follows a normal distribution.

c) The stock price S follows a lognormal distribution.

d) This model does not impose mean reversion.

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FRM EXAM 2006—QUESTION 87. Which of the following is true about stress testing?

a) It is used to evaluate the potential impact on portfolio values of unlikely, although plausible, events or movements in a set of financial variables.

b) It is a risk management tool that directly compares predicted results to observed actual results. Predicted values are also compared with historical data.

c) Both a. and b. are true.

d) None of the above are true.

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FRM EXAM 2007—QUESTION 101. A large, international bank has a trading book whose size depends on the opportunities perceived by its traders. The market risk manager estimates the one-day VAR, at the 95% confidence level, to be USD 50 million. You are asked to evaluate how good a job the manager is doing in estimating the oneday VAR. Which of the following would be the most convincing evidence that the manager is doing a poor job, assuming that losses are identical and independently distributed (i.i.d.)?

a) Over the past 250 days, there are eight exceptions.

b) Over the past 250 days, the largest loss is USD 500 million.

c) Over the past 250 days, the mean loss is USD 60 million.

d) Over the past 250 days, there is no exception.

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FRM EXAM 2009—QUESTION 5-8. Greg Lawrence is a risk analyst at ES Bank. After estimating the 99%, one day VAR of the bank's portfolio using historical simulation with 1,200 past days, he is concerned that the VAR measure is not providing enough information about tail losses. He decides to reexamine the simulation results. Sorting the simulated daily P&L from worst to best gives the following results:

Rank P&L	1 −2,833	2 −2,333	3 −2,228	4 −2,084	4 −1,960	6 −1,751
Rank P&L	7 −1,679	8 −1,558	9 −1,542	10 −1,484	11 −1,450	12 −1,428
Rank P&L	13 −1,368	14 −1,347	15 −1,319			

What is the 99%, one-day expected shortfall (ES) of the portfolio?

a) USD 433

b) USD 1,428

c) USD 1,861

d) USD 2,259

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FRM EXAM 2009—QUESTION 5-14. Which of the following statements about expected shortfall (ES) is incorrect?

a) ES provides a consistent risk measure across different positions and takes account of correlations.

b) ES tells what to expect in bad states: It gives an idea of how bad the portfolio payoff can be expected to be if the portfolio has a bad outcome.

c) ES-based rule is consistent with expected utility maximization if risks are ranked by a second-order stochastic dominance rule.

d) Like VAR, ES does not always satisfy subadditivity (i.e., the risk of a portfolio must be less than or equal to the sum of the risks of its individual positions).

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EXAMPLE 1.4: FRM EXAM 2009—QUESTION 1-6

Suppose Portfolio A has an expected return of 8%, volatility of 20%, and beta of 0.5. Suppose the market has an expected return of 10% and volatility of 25%. Finally, suppose the risk-free rate is 5%. What is Jensen's alpha for Portfolio A?

a. 10.0%
b. 1.0%
c. 0.5%
d. 15%

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EXAMPLE 5.3: FRM EXAM 2002—QUESTION 3

Consider a stock with daily returns that follow a random walk. The annualized volatility is 34%. Estimate the weekly volatility of this stock assuming that the year has 52 weeks.

a. 6.80%

b. 5.83%

c. 4.85%

d. 4.71%

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EXAMPLE 6.3: FRM EXAM 2002—QUESTION 51

Consider a savings account that pays an annual interest rate of 8%. Calculate the amount of time it would take to double your money. Round to the nearest year.

a. 7 years

b. 8 years

c. 9 years

d. 10 years

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EXAMPLE 6.2: FRM EXAM 2009—QUESTION 4-9

Lisa Smith, the treasurer of Bank AAA, has $100 million to invest for one year. She has identified three alternative one-year certificates of deposit (CDs), with different compounding periods and annual rates. CD1: monthly, 7.82%; CD2: quarterly, 8.00%; CD3: semiannually, 8.05%; and CD4: continuous, 7.95%. Which CD has the highest effective annual rate (EAR)?

a. CD1

b. CD2

c. CD3

d. CD4

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13-1. Which currency pair would you expect to have the lowest volatility?

a) USD/EUR

b) USD/CAD

c) USD/JPY

d) USD/MXN

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EXAMPLE 5.5: FRM EXAM 2004—QUESTION 39

Consider a portfolio with 40% invested in asset X and 60% invested in asset Y. The mean and variance of return on X are 0 and 25 respectively. The mean and variance of return on Y are 1 and 121 respectively. The correlation coefficient between X and Y is 0.3. What is the nearest value for portfolio volatility?

a. 9.51

b. 8.60

c. 13.38

d. 7.45

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- A fund manager announces that the fund's one-month 95% VaR is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?
- A fund manager announces that the fund's one-month 95% expected shortfall is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?

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- Suppose that the change in the value of a portfolio over a one-day time period is normal with a mean of zero and a standard deviation of $2 million; what is

  a. The one-day 97.5% VaR

  b. The five-day 97.5% VaR, and

  c. The five-day 99% VaR?

- What difference does it make to your answer to last problem if there is first-order daily autocorrelation with a correlation parameter equal to 0.16?

Tarea larga 1

- Suppose that we back-test a VaR model using 1,000 days of data. The VaR confidence level is 99% and we observe 17 exceptions. Should we reject the model at the 5% confidence level? Use a one-tailed test.
- The change in the value of a portfolio in one month is normally distributed with a mean of zero and a standard deviation of $2 million. Calculate the VaR and ES for a confidence level of 98% and a time horizon of three months

Administración actuarial de riesgos financieros

Tarea larga 1

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